JEFECORP

JefeCheck

BETA

PLAY.PROCESS.SHARE

(WORK IN PROGRESS, FINAL VERSION TO BE RELEASED WITH SOFTWARE RELEASE)

User's Manual

JEFECORP

JefeCheck User's Manual

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Foreword

Thank you for buying or evaluating JefeCheck from Jefecorp! This user's manual will help you understand what JefeCheck is all about and have you using it like a pro in no time.

The manual is divided in three sections, for beginners, check out **Section 1: Play**. To use the more advanced features in JefeCheck look at **Section 2: Process** and **Section 3: Share**.

Also please note that this manual is a work in progress for the Beta Version of JefeCheck. The final user's manual with all the sections complete will be released along with the final release of the software.

Have Fun!

Daniel Gollás, Developer of JefeCheck

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Introduction

What is this software for anyway?

CheckMate is targeted at many different people inside the film and TV industry. Particularly those working with **frame based data**.

The advantages of working with frame based data are many, but a big disadvantage is the fact that there is no straight forward way of playing them back in sequence in their native format. A jpeg for example has no idea that it belongs to a series of files.

That is where image sequence players come into the game. But JefeCheck is a lot more than just an image sequence player!

JefeCheck has three big features, that we call **Play**, **Modify** and **Share**.

Play: The Core of JefeCheck. Load up an image sequence and **play it back at full resolution**. Load more, than one, load up to four. Look at them, compare them, move them around, look closer, play them faster, play them slower, scrutinize every frame to make sure you are looking at exactly what you expected to see.

Process: The first thing that sets JefeCheck apart from the rest. Process your images (non destructively of course) in any way you can think of with a powerful, easy to use plug-in system based on a well known shading language. Apply 3D Lookup Tables, adjust brightness, contrast, saturation and gamma, composite one sequence over another, do a preview green screen extraction, apply primary color correction, create difference mattes etc…. **All in real time and in full resolution**!

Share: So you're having fun playing back and modifying your images, but playing alone is no fun. Fortunately **JefeCheck is multiplayer**! Start a remote session with any other JefeCheck enabled computer over a local area network or over the Internet[1][2] and play together. It's like being in your own virtual screening room. Everything you do is mirrored on their JefeCheck and vice versa. Text Chat, plug-in and Lookup Table sharing, remote pointers and other tools help you show everybody in the remote session exactly what you want them to know.

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[1] All participants in a remote session need to have the images stored locally, no image information travels over the Internet, so all your information is safe.

[2] At least one of the participants over an Internet remote session must have a public IP address.

What JefeCheck is NOT:

JefeCheck offers a lot of image processing power to alter an image in real time, but JefeCheck is not a finishing tool. Although you can use JefeCheck to review the final output of your work, you should not rely on its image processing features to complete your work. Although it is indeed possible to render out Pre-composites and basic color correction, you should only use these tools as means to a preview. For example, you can do primary color correction to your work to get an idea of changes that you would like to make using programs that you might not have immediate access to.

Using the remote collaboration capabilities of JefeCheck, someone a long distance from you might be able to give you an idea of changes that they would like to make to your sequences, but for final approval of such delicate subjects such as color correction, there is really no comparison to having a real live meeting where both people are looking at the exact same monitor with the exact same color temperature etc.

How to read this manual

This manual is written in a way that you can read it cover to cover, but if you need to know about something in particular you can just jump any of the three sections (Play, Process or Share).

It is recommended that you understand the sections in order, since each one assumes you are familiar with the terms described in the previous one.

ICON KEY
✎ Summary
⌨ Shortcuts
✓ Power Tip

You will also see some paragraphs that contain special information, identified with a special icon. These paragraphs usually give extra tips or a short summary of what was just discussed. The icon key on the left shows what each icon means.

There is also an index if you need to look up a particular concept.

Finally, there is an appendix, which shows you how to build custom plug-ins for JefeCheck.

PLAY

1

SECTION 1: PLAY

In this section you will learn the basics of JefeCheck.

At its core JefeCheck is an image sequence player, that is, it allows you to load a bunch of images, named in a sequenced fashion (image.0001.jpg, image.0002.jpg ... etc) and play them back at a given frame rate.

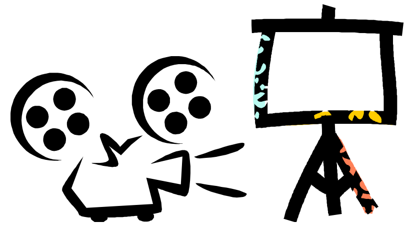
JefeCheck allows you to load up to **four different** image sequences, and then watch them on screen in several different ways.

Tracks and Viewports

There are two basic concepts you need to grasp to use JefeCheck: Tracks and Viewports. Tracks are where you load the sequences, and Viewports are where the images are actually drawn for you to see.

Although they are closely related, Tracks and Viewports are independent of each other and allow a great deal of flexibility. In a movie theater analogy, you can think of a Track as an actual film reel and a Viewport as the combination of projector and screen: A reel will always contain the same images, but depending on variables like the size of the screen or the lens on the projector, you will see the image in different ways. In JefeCheck, a Track can be seen in many different ways, simply by adjusting the properties of the Viewport on which you are looking it at.

Track



Viewport

You can load up to 4 different tracks at once (Tracks A, B, C and D), and then watch them on up to four different viewports (Viewport 1, 2, 3 and 4).

So to sum it up...

You load sequences into tracks, and then you watch those tracks on one or more viewports.

The most basic way of using JefeCheck is to simply see one Track on one Viewport without changing anything else. To do this, let's first get acquainted with the basic user interface.

When you first open JefeCheck, you will see the **Main Window** and the **Load Window** on top.

If you don't see the Load Window, you can bring it up by pressing the shortcut *Ctrl-L* (L is for load).

Basic User Interface

When you first open JefeCheck, you will see the **Main Window** and the **Load Window** on top. If you don't see the Load Window, you can bring it up by pressing the shortcut *Ctrl-L* (L is for load). As you might have guessed, the Load Window is

used to load the image sequences into tracks.

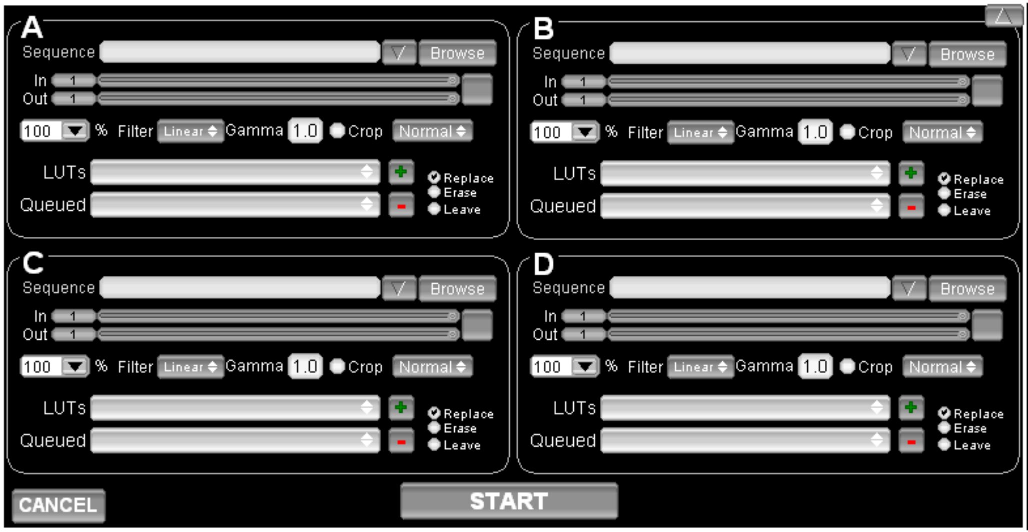


Figure 1 The Load Window

The Load Window

The load window is shown on **Error! Reference source not found.**.

The load window is divided into 4 sections. These four sections represent the four tracks into which you can load image sequences. You don't have to use all four, you can use as many as you want. Each section is exactly the same, so let's analyze what each control in the section does.

Figure 2 "Loading Parameters"

Sequence Info Controls

The Sequence Info Controls let you select what sequence you want to load. When you press the **Browse Button** a file choosing dialog appears as shown in Figure 3.

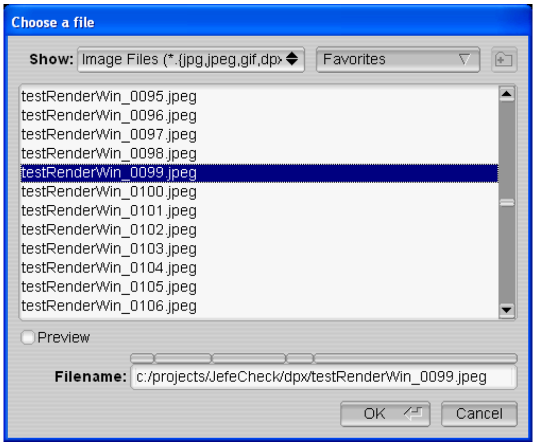


Figure 3 The file chooser dialog

From within the file chooser, you can navigate through all the folders and drives attached to your computer. To pick a sequence simply select any file that belongs to that sequence and click OK.

You can filter out the kinds of files that are shown by selecting an item in the **Show** selection box.

The bottom text box labeled **Filename** shows the current path to the selected file, and also serves as a navigation tool, click on the box above any folder in the path to go directly to that folder.

When working on a project, you will probably be visiting the same folder many times a day. JefeCheck lets you store **Favorites**, which are paths to folders you use frequently. To add the current folder to your favorites click on the Favorites selection box and choose **Add to Favorites**. Now that folder will be available from the Favorites selection box and you can navigate directly to it simply by choosing it. You can also manage your favorites by clicking on the Favorites selection box and choosing **Manage Favorites**.

Once you have selected the file you want, you click OK and the file chooser dialog closes. And the load window is populated with information relevant to the sequence you have chosen as shown in Figure 4.

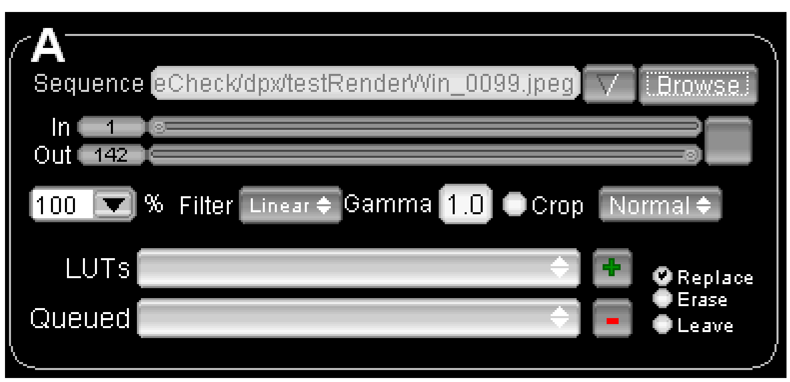


Figure 4 Populated Sequence Options

The most recent sequences you load will be available from the **Recently Loaded** drop down menu to the left of the browse button.

Even though you are not familiar with the Main Window yet, you will see that the image you selected will be loaded and previewed on it, right behind the Load Window.

You can get a better look at the image behind by moving the Load Window out of the way. You can **click and drag on any black part of the window to move it**. You can also **shrink the load window by clicking on the little arrow button** on the top right corner. You **restore the load window by clicking on the little arrow again.**

On top of that preview image you will also see some information on the sequence you are going to load, such as the filename, format, number of channels, bits per channel, and how many frames will be loaded as shown in figure.

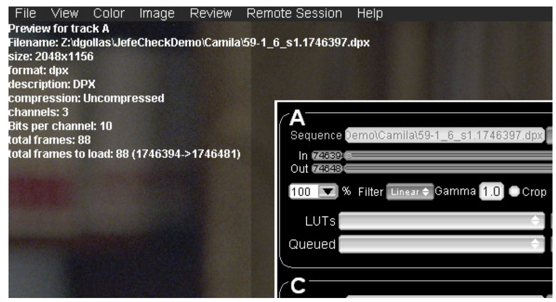


Figure 5 Preview information

At this point, you could press the **START Button** and JefeCheck would begin to load the sequence into the track for you to play.

But before we do that, let's review the other options you have at load time.

Range Controls

The Range Controls let you select what part of the whole sequence you want to load. You simply select **In** and **Out** frames. The numbers you see are directly related to the numbers on the file's name. So if a sequence starts with frame number 153, the smallest value you could choose for the in frame, would be 153. To select in and out points, you simply drag the sliders on the right side of the numbers.

If you press the Reset Button on the right side of the sliders, they will return to their default positions at the beginning and end of the sequence. When this happens, the sequence is re-analyzed and if any new frames have appeared in the sequence they will be taken into account (maybe they just got rendered or copied to the folder).

Load Time Modification Controls

The Load Time Modification controls allow you to alter certain properties of the image sequence as it is loaded into a track. Changes made at load time are "baked" into the track, as opposed to **Real Time Modifications** which are applied "on the fly" as the picture is being drawn on the screen (you will learn about Real Time Modifications later).

Let it be clear that neither Real Time Modifications nor Load Time Modifications change the original file on disk in any way.

A comparison between Load Time Modifications vs. Real Time Modifications is shown in Table 1.

Table 1 Load Time vs. Real Time Modifications

Load Time	Real Time
Loading a scaled image saves memory.	**A scaled image uses the same amount of memory as a full resolution one.**
Changes are baked in and can't be reversed unless sequence is reloaded.	**Changes are done to the sequence on the fly, and can be turned on and modified instantly.**
Loading takes longer	**Loading is faster**
Works well on old hardware	**Runs best on newer hardware**

Not all Load Time Modifications have a Real Time counterpart, and therefore using one type of modification should not exclude using the other. This will become clearer when we start to use and understand Real Time Modification Controls.

The Load Time modification controls are described below:

Scale

The scale input box allows you to load the images as proxies of the original one. This means that the image is resampled and loaded as a lower resolution version of itself. This saves memory and also helps performance on lower end hardware. You can choose from common options by clicking on the pop down menu (100%, 50%, 25%) or type in any number. When you modify the Scale value, the preview image in the main window changes accordingly.

Filter

This option box lets you choose between two resampling methods for when scaling an image, if you are loading the image at 100% scale, this option is irrelevant. Linear sampling is faster, but produces lower quality images, while Bilinear sampling is slower, but produces smoother images. If you are not that concerned with the quality of the proxies and want faster load times use Linear sampling, otherwise, choose Bilinear.

Gamma

You can modify the gamma of an image at load time by changing the value on the gamma slider to something other than 1. When you change the value of the slider, the preview image in the main window changes accordingly.

Changing the gamma on an image at load time takes time, and if your hardware allows it, you should use the Real Time alternative to this (see Section 2: Processing and Appendix for Plugins.).

Crop

Crop is a load time modification that has no Real Time counterpart, and is great for saving memory and improving performance on lower end computers.

Crop allows you to specify a rectangular sub-section of the image you want to load, and only that part will be loaded into memory. When you enable the **Crop Checkbox** a blue overlay square will appear atop the preview image in the main window. Whatever is under this square will be loaded. **To move the Crop square, click and drag with the right mouse button, to resize the square, shift-right click and drag**.

Format

The format drop down box allows you to choose different formats to load the image into memory. The different formats have different tradeoffs as listed:

- Normal – This loads the image and transforms it to an 8 bit per component image to be displayed on normal monitors. This is the format you will usually use, as it offers a good balance of memory usage and provides perfect image quality unless heavy processing is applied later.

- S3TC – This loads the image and compresses it in order to fit more frames into memory. Not all hardware supports this format and it is a little slower to load. The compression is not lossless and introduces a small amount of artifacts, so if you are checking for pixel imperfections, this is not the way to go. On the other hand, this format let's you fit up to three times more frames into memory.

- Depth Reduction – This format has a limited amount of uses since it loads the

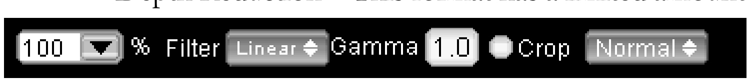


  image and transforms it into a 4 bits per component image. This introduces serious banding and color inaccuracies for all but the simplest images. The tradeoff is that you can fit twice as many frames into memory as you could with the Normal format. One of the uses of this format is to load as many frames into memory in order to check for black or missing frames in a sequence, a process where image quality is not important.

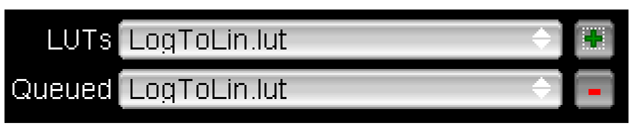
- Half Float – This image format loads images in their native bit depth and transforms them to 16 bit floating point (half float) images. This uses up twice the amount of memory than the normal format, but allows greater flexibility during processing, preventing banding and other color artifacts when applying real time color modifications.

Unless you are having specific problems, or you need to load more frames into memory at the expense of image quality, **the Normal method is the recommended format for most uses**.

Load Time LUT Controls

JefeCheck allows you to apply 1D and 3D Lookup tables to an image sequence in order to match a certain calibration or color correction attributes. A basic and commonly use of a LUT is to convert an image from Logarithmic Color Space into Linear Color Space (or vice versa).

With JefeCheck you are not limited to one LUT, you can apply and layer one LUT on top of another, as many times as you need. As an example, this is useful to get an idea of how an image sequence will look when printed to film, going through a series of processes like printing to negative, then to positive, and finally projecting with a specific kind of projector. If you have a LUT simulation of each one of these processes, you can layer one on top of the other easily.



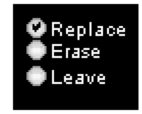
At load time, you use the LUTs Dropdown Menu to select from any available LUTs loaded into JefeCheck (To learn how to load LUTs see Section 2: Process) and then click the **Green Plus Button** to add the selected LUT to the queue of LUTs that will be applied to the image. The order in which you add the LUTs will be the order in which they will be applied to the image.

As you add LUTs with the **Green Plus Button**, the **Queued Dropdown Menu** will fill up with the LUTs you add. To remove a LUT from the queue, selected from the Queued Dropdown menu and click the **Red Minus Button**. You can add as many or as few (or none) LUTs to the queue.

Applying LUT at load time load makes the loading process considerably slower. If your computer and graphics card support it, you should opt for Real Time LUT application, as you will learn in Section 2: Processing. The Load Time LUT application feature is used mainly in lower end systems that need the color accuracy but lack the modern graphics hardware to do it in Real Time.

Track Loading Behavior

In JefeCheck you can load up to four different sequences, but you don't have to load them all at the same time if you don't want to. Maybe you load up a sequence and then you remember that you also wanted to load another one into another track. Of course, when you load the second sequence, you probably don't want to load the first one again, or maybe you want to replace it with a completely different sequence. The Track Loading Behaviors allow you to control all this.



You can choose any of the three behaviors to tell the track what you want it to do when you click the START Button.

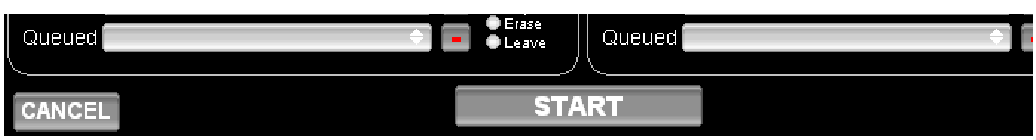
- Replace – The default option, it simply unloads whatever is already loaded into the track, and loads the sequence specified in the sequence controls into that space.

- Erase – This options unloads whatever is loaded in the track and does nothing else. This frees up memory in case you don't need to see the sequence any more. (**This does not erase the sequence in disk, it simply unloads it from memory**).

- Leave – This option does nothing to the track. If there is a sequence there, it simply leaves it there. This is useful when you already loaded a sequence into a track, and you don't want to reload it when loading a second sequence into another track (or third or fourth).

START and Cancel Buttons

Once you have selected what tracks you want to load, adjusted their load time parameters (if needed) and you are sure of what you want to load, you hit the **START Button** on the Load Window. This closes the Load window and the sequences will start loading into the tracks. If you decide you don't want to do anything, you can simply click the **Cancel Button** to close the Load Window**.**



So to sum it up...

You use the load window to select what tracks you want to load, adjust how you want them to be loaded, and then click the START button to start the loading. All parameters have the most common values as default, so loading can be as fast as selecting a sequence and clicking START.

The Main Window

The main window is where almost everything else in JefeCheck really happens. It is basically divided in three, the Menu Bar, the Viewports Area, and the Control Bar as you can see in **Figure 6**.

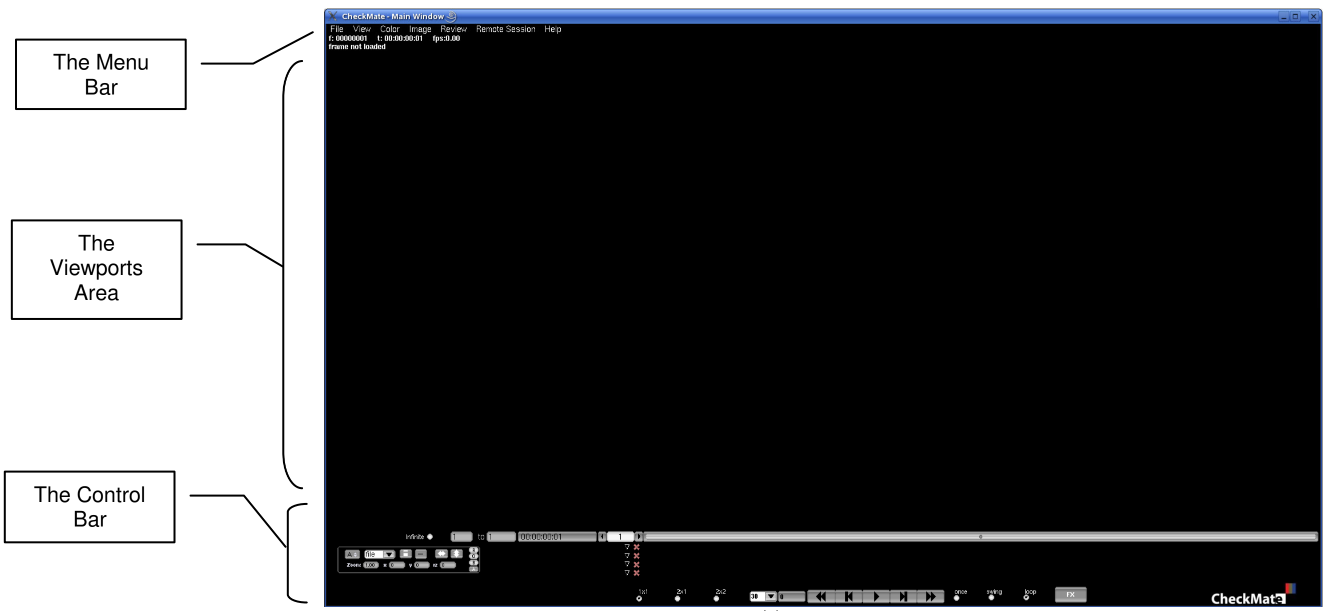


Figure 6 "The Main Window".

- **The Menu Bar** is where you access other windows or adjust some settings.
- **The Viewports Area** is where the viewports live and you actually see the image sequences you have loaded.
- **The Control Bar** is where you control almost everything related to the playback of the sequences you are watching and adjust many setting for the viewports.

You can hide the Control and Menu Bars by pressing the **h Key**. You bring the controls back up by pressing **h** again.

When you first see the Main Window, it is empty. What is shown in the viewports area varies depending on what windows are open, but this is the area where you will see all of the graphical data that JefeCheck can show.

When the Load Window is open, you will see a preview of the sequences that will be loaded if you click the START Button. When the Load Window is closed, you will see whatever track is loaded and assigned to each particular viewport.

Remember tracks are independent from viewports: tracks hold sequences, and viewports are used to view those sequences. One viewport can show only one sequence, but one sequence can be seen on multiple viewports.

Let's go over each element of the Main Window.

The Control Bar

Figure 7 Shows the Main Window Control Bar, which contains many different controls that can grouped into three logical areas as shown in **Error! Reference source not found.**.

Figure 7 The Main Window Control Bar

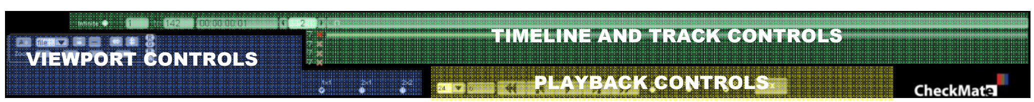


Figure 8 The three big logical areas of the control bar

The three big logical areas in the control bar are:

- Timeline and Track controls
- Viewport Controls
- Playback controls

Timeline and Track Controls

The Timeline and Track Controls show you information relevant to time and position of what you are seeing on the screen and they let you control what frames of the loaded sequences you are looking at.

1 2 3 4

Figure 9 Timeline Controls

The Timeline controls shown in Figure 9 from left to right are as follow:

1. **From-To Markers**: The **From** and **To** markers specify the range of the timeline. You can force the timeline to playback only from a certain frame and up to another one. This is helpful if you only want to see a certain range of frames of the sequence at a time. This is different from only loading a certain range of frames, since you can change the From and To markers on the fly without having to unload or reload any frames. As with all numeric inputs in JefeCheck, you can click and drag to change the value or enter a number with the keyboard.

   Whenever you load a sequence, the From and To frames are updated in order to fit the longest loaded sequence.

2. **SMPTE Counter**: The SMPTE Counter shows the current time relative to the timeline starting at frame one, it does not show time code information embedded in files. The Counter shows time in hours, minutes, seconds and frames in the following way: HH:MM:SS:FF. The amount of frames that have to elapse for the Seconds counter to increase depends on the target frame rate, controlled from the Playback Controls (see Playback Controls)

3. **Current Frame**: Shows the current frame the timeline is at.

4. **The Timeline**: The timeline allows you to control and graphically visualize what frame of the sequence you are looking at. You can click and drag the knob on the timeline to scrub or jump to any point on it. The size of the knob will vary depending on the range the timeline is showing, defined by the From and To markers.

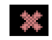
The Track Controls are shown in Figure 10 along with the Timeline.

Figure 10 The Track Controls with the Timeline

You can see that there are four rows of controls under the Timeline. These four rows represent the four tracks that can be loaded. Figure 10 shows a single loaded sequence. Each row contains a **Track Options Button**, a **Cancel button** and a **Track Bar.**

1. **The Track Bar**: 

The Track Bar consists of a gray background that shows how long the track is in relation to the timeline and other tracks. Inside the gray background, there is a green stripe which shows what frames are loaded in memory and ready to show. While a track is loading, you can playback whatever part is already in memory.

2. **The Cancel Loading Button**: 

The Cancel Loading Button turns red whenever its corresponding track is being loaded. This means that you can cancel the load at any point. When the track is completely loaded or no more frames can fit into memory, the loading stops and the Cancel Loading Button turns gray.

3. **The Track Options Button**: 

When you click on the Track Options Button a Track Options window appears as shown in Figure 11

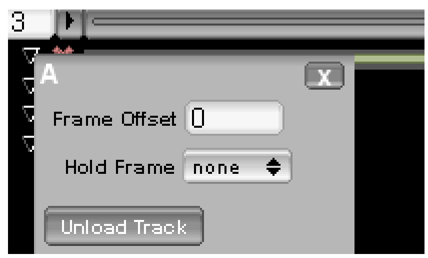


Figure 11 The Track Options Window for track A

The Track Options window shows several options for track behavior.

- **Frame Offset**: Allows you to offset the track a certain amount of frames in respect to the timeline and other tracks. This is useful if you are comparing two tracks but the segment that you wish to compare is not necessarily in the same frames on both tracks. For example, if you want to compare and simultaneously play back the last 10 frames of Track A and the first 10 frames of Track B you could offset track B enough frames so that the first 10 frames align with the last 10 frames of Track A. Or you could offset track A in the reverse direction to obtain a similar effect.

- **Hold Frame**: Many times, you want to simultaneously play back two sequences of different lengths. The Hold frame specifies what should be displayed if the current frame is outside of the loaded frames for a particular track. For example, if Track A is 25 frames long, and track B is 45 frames long, when the timeline is at frame 30, track B will show its own frame 30, but track A will show a black frame since there is nothing loaded there. The Hold frame can be one of three values

  - First Frame: Whenever the current timeline frame is smaller than the first loaded frame of the Track, the first loaded frame will be shown.

  - Last Frame: Whenever the current timeline frame is larger than the last loaded frame of the Track, the last loaded frame will be shown

  - Current Frame: No matter what value the current timeline frame is, the Track will show the frame position that was current when the **Current Frame** option was selected. So if the timeline value is 5, and you select the Current Frame option, track A will always show frame 5.

- **Unload Track**: You can unload a particular track in order to free memory or simply because you don't want to use it anymore. When you click this option you can simply unload the sequence, or unload the sequence and erase all settings from the Load Window regarding this track.

Many times a whole sequence won't fit into memory. You can tell JefeCheck that you want to unload the loaded frames and start loading the sequence from a different point by **alt-right mouse button** clicking on the timeline on the point that you would like to start loading. All tracks will unload and start loading from that point on.

So to sum it up…

You use the Timeline controls to change timeline range and current timeline frame, which controls what is displayed for each track according to its own settings.

Playback Controls

The Playback Controls let you control and decide how the timeline is to behave.

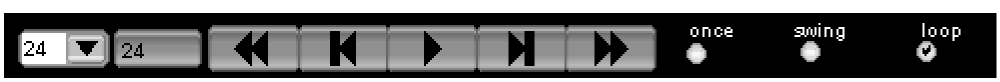


Figure 12 The Playback Controls

An explanation of each control (from left to right) follows:

1. **Target Frame rate**: When you playback a sequence, JefeCheck will try to maintain the target frame rate. You can type in any integer value or click on the drop down menu arrow to select commonly used frame rates.

2. **Actual Frame Rate**: This control shows the actual frame rate the sequence is running at. Due to hardware limitations the playback frame rate may not match the target frame rate.

3. **Rewind**: Returns the current timeline value to its first frame. The keyboard shortcut is **M** .

4. **One Frame Back**: Turns the current timeline value one frame back. The keyboard shortcut is **< or . (period)**.

5. **Play/Pause**: Starts/Pauses playback. When JefeCheck is playing, the label on the button changes to a pause symbol and the same button pauses playback. The keyboard shortcut is **Space Bar.**

6. **One Frame Forward**: Turns the current timeline value one frame forward. The keyboard shortcut is **> or , (comma)**.

7. **Fast Forward**: Turns the current timeline value to its last frame. The keyboard shortcut is **? or /**

The keyboard shortcuts for the playback controls may seem a little weird, but if you **position your right hand with your thumb on the space bar, and your index finger on the M key, you will see that your other fingers will fall right in place with the other playback control keys**, allowing you to start/pause playback, fast forward, rewind, advance and recede frames with very little effort in a very intuitive way.

8. **Loop Modes**: The loop modes control what happens when the current timeline value reaches its limit.

   - Once: The sequence plays once and then stops on the last frame. Keyboard shortcut **8**.

   - Swing: The sequence plays in the forward direction, and when it reaches the last frame, it starts playing in reverse until it reaches the first frame, going back and forwards until it is paused. Keyboard shortcut **9**.

   - Loop: The sequence plays in the forward direction until it reaches the last frame at which point it starts over at the first frame, in an infinite loop until it is paused. Keyboard shortcut **0**.

So to sum it up...

You use the Playback controls to start/pause the playback, determine the target frame rate, monitor the actual frame rate and select if and how you want the sequence to loop.

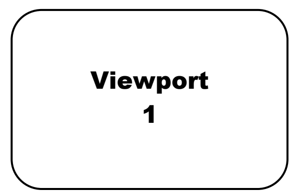
Viewport Controls

The viewport controls, shown in Figure 13 allow you to specify and control what viewports will be shown, and what track each viewport will display.

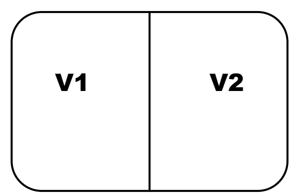
Figure 13 The Viewport Controls

The checkboxes on the right, represent commonly used viewport configurations or Layouts. But you can access more configurations from the keyboard or in the View>Layout menu in the Menu Bar.

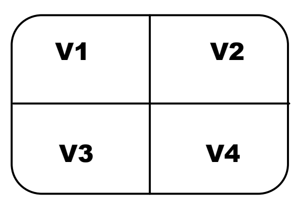
- **1x1**: A single viewport (Viewport 1), using up all the space in the viewport area is shown. Keyboard shortcut **1**.



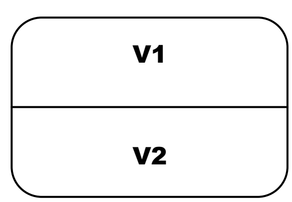
- **2x1**: Two viewports are shown side by side (two columns times one row), each using half the horizontal space and the full vertical space in the viewport area. Keyboard shortcut **2**.



- **2x2**: Four viewports are shown, two side by side and another two under them (two columns times two rows), each using up half the vertical space and half the horizontal space in the viewport area. Keyboard shortcut **3**.



- **1x2**: Two viewports are shown one on top of the other (one column times two rows), each using



half the vertical space and the full horizontal space in the viewport area. Keyboard shortcut **4**.

Figure 13 Shows the Viewport controls when all four viewports are showing in a **2x2 layout**. When less viewports are visible, their controls hide away to avoid clutter.

As you can see, each viewport has an identical group of controls that mirrors its position on the viewport area. Since every viewport control group is the same, we will focus on a single one as shown in Figure 14.

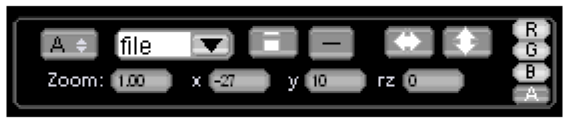


Figure 14 A Single Viewport Control Group

Each control group affects how each viewport is displayed. A description of what each control does follows:

- **Track Selection Box** :
  Select which track you want displayed on this viewport. It can be any of the four tracks. If no sequence is loaded in the selected track, the viewport will not show anything.

  More than one viewport can show the same Track at the same time, so you could see the whole frame in one track, while looking at a close up of the same frame in another viewport.

- **Aspect Ratio** :
  You can change the aspect ratio of the sequence being shown on the screen in real time. The default value for the Aspect Ratio control is **file**, which means that the original aspect ratio of the image being shown will be used to display it. The drop down menu options will show commonly used aspect ratios such as 4:3, 16:9, 2.35:1 etc, **but you can type in any ratio you want, either in the x:y notation, or as a decimal number obtained by dividing x/y.**

  By default, when you change the aspect ratio of a viewport, the image will be deformed to adapt to it. This is useful when viewing material filmed with anamorphic lenses. If you don't want to deform the material but still want to change the aspect ratio, you use **Crop Bars control**.

- **Crop Bars control** **:**
  When you activate the Crop Bars control and the Aspect Ratio control is set to

anything other than "file", the aspect ratio of the image will be changed not by squashing or stretching the image, but by applying black crop bars to hide or compensate a portion of the image.

- **Crop Bars Positioning :**
  You can control how to crop the image when Crop Bars are enabled. By default, two equally sized crop bars appear over the image, on at the top and one at the bottom. Each time you click the Crop Bars Positioning control you toggle through one of three modes: Centered (the default), Top (a single, double sized crop bar is placed on the top part of the image) or Bottom (a single, double sized crop bar is placed on the bottom part of the image).

- **Flip/Flop Controls :**
  When you activate the flip/flop controls, the viewport will reflect vertically (flip) or horizontally (flop) or both.

- **RGBA Masks :**
  You can see any RGB color channel (or any combination) of the image in the viewport by clicking the RGB color masks on or off. So if you only want to see the Red and Green Channels, you can turn of the Blue channel by clicking on the B button. If the sequence loaded in the track being shown by this viewport has an Alpha Channel it can be shown/hidden by turning on/off the A Mask. If no alpha channel is present the frame will be drawn white.

- **Zoom Control :**
  In JefeCheck you can interactively zoom in and out of a viewport by using the scroll wheel on the mouse, or by using the zoom control. A value of 1 means no zoom. As with other numeric controls in JefeCheck, you can click and drag your mouse to modify the value on it. The rate of change when dragging will depend on what button you click with. **You can also zoom in and out by holding down the Z key on the keyboard and right-clicking and dragging on the viewport**.

- **X/Y position :**
  In JefeCheck you can interactively pan a viewport by left-clicking and dragging the mouse on the viewport, or by using the x/y position controls. X/Y controls with a value of 0 will result in a viewport centered within its allotted viewport area.

- **Rotation Control :**
  In JefeCheck you can interactively rotate a viewport as if a pin was put through the middle of the image by using the rotatation control.

So to sum it up...

You use the layout controls to change how many viewports are displayed and how they are laid out on the viewports area of the Main window.

You use the viewport controls to change the way each viewport is drawn, changing parameters like the aspect ratio and zoom values and what track is displayed on each viewport.

The Viewports Area

The central area of the Main Window is the Viewports Area. This is where the image sequences and other bits of information are drawn. The viewport area will be divided into up to four sections, depending on the layout that has been chosen (see Viewport Controls on page 18).

The sections are not divided by a visible divider, but you can see that each viewport also has a text overlay that shows some valuable information on what is being shown at the moment.

Viewport Behavior

Viewports show different information depending on what you are doing at the moment. Most of the time you will see images, and sometimes you will see other kinds of information, like the graphical representation of a Lookup Table.

The basic situations where you will see different things on the viewport are the following:

- **With the Load Window Open**

  When the load window is open, each one of the visible viewports (remember you can change what viewports are visible by changing the layout, see Viewport Controls p.18) shows a preview frame for each sequence that will be loaded into each track. **The preview for track A will be shown on Viewport 1 (V1), track B on V2, track C on V3 and track D on V4.** This is the only time that the viewports are fixed to a particular track, so you will always see the preview for each track on the assigned viewport, Notice how the different sections for each track mirror the layout of the viewports showing their preview: the load setting for track A are on the top left corner, so is the viewport showing it's preview.

  Additionally, when the Load Window is open, the text overlay on each viewport will show relevant information regarding the sequence that will be loaded into its corresponding track as shown in Figure 15. The text information you will see is:

- What track the preview is for.
- The filename of the image being displayed, which is also the one from where the sequence of images will be derived
- The image's size in pixels (Width x Height)
- The image's format.
- A description of the format (if available).
- The image's compression method (if any).
- How many channels are in the image.
- How many bits per channel the image has.
- The total number of frames identified in the sequence.
- The total number of frames that will be loaded, followed by the start and end frame numbers according to the files numbering (The total number of frames to load depends on the In/Out track loading controls, see **Range Controls** p7).

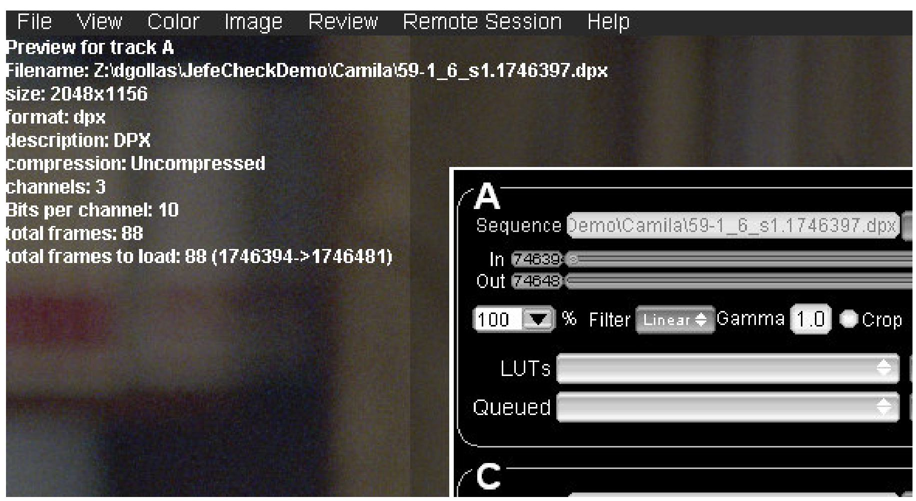


Figure 15 Preview text for track A

- **During Playback**

  Once you have sequences loaded into tracks and the Load Window is closed, you can see the tracks on any viewport you want, in any layout you want.

Tracks will be displayed on whatever viewports have them assigned, and a text overlay will show relevant information. The text can be turned on or off by pressing the shortcut key **t**. If the track contains a DPX file format sequence, pressing **shift+t** will toggle the display of the file's metadata. Pressing **alt+t** will toggle the text display for all viewports.

The information that will be displayed when the text overlay is enabled as seen in Figure 16 is as follows:

- The current timeline frame (eg. *f:0000086*)
- The current SMPTE time relative to the first frame of the timeline (eg. *t:00:00:03:14*).
- The current playback framerate (eg. *fps:24.00*)
- The filename of the image being shown.
- The image resolution, the number of channels, and the number of bits per component of the displayed image (eg. *2048x1156x3 10bpc*)

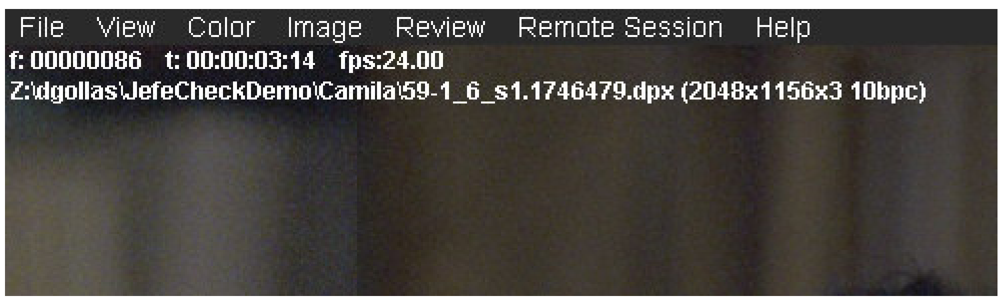


Figure 16 Playback text overlay on the viewport

When DPX metadata is present and is enabled, the information that will be displayed as seen in Figure 17 is the same as normal frames, plus the following fields:

- Timecode (from the DPX TV Header)
- KeyKode (derived from the DPX Motion Picture Header)
- Filename (from DPX File Information Header)
- Source Filename (from DPX Image Orientation Header)
- Frame ID (from DPX Motion Picture Header)
- Filename (from DPX File Information Header)

- Project (from DPX File Information Header)
- Slate (from DPX Motion Picture Header)
- Creator (from DPX File Information Header)
- Input Device (from DPX Image Orientation Header)
- Input Serial (from DPX Image Orientation Header)
- Input Device (from DPX Image Orientation Header)
- File Size in MB, KB and Bytes (from DPX File Information Header)
- Frame rate (from DPX Motion Picture Header)
- Format (from DPX Motion Picture Header)
- Creation Time (from DPX File Information Header)
- Source Creation Time (from DPX Image Orientation Header)
- Copyright (from DPX File Information Header)

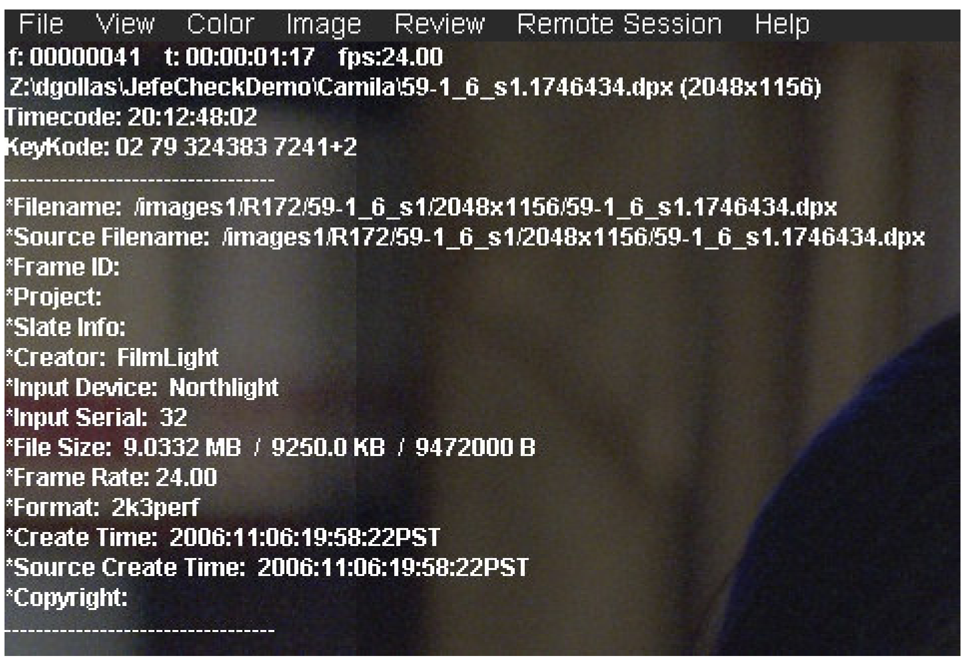


Figure 17 DPX Metadata Text Overlay

- LUT visualization

The viewports are not only useful for displaying images, they can also show a graphical representation of a 1D o 3D Lookup Table. Since we haven't covered loading LUTs and the LUT Load Window, we will not go into greater depth, but you can get more information in Section 2: Process.

Viewport Interaction

Using the mouse and keyboard, you can interact with the viewports to modify their parameters (see Viewport Controls p18) and obtain information about what is being displayed.

A list of what you can do with a viewport using the mouse and keyboard is presented here.

To do this…	…do this
Zoom in and out	Move the mouse wheel up or down. Pressing **Shift** makes the zoom act 10x faster. Optionally, Right-click and drag while holding the **Z** keyboard key.
Zoom in and out of all viewports at the same time (Gang Zoom)	Move the mouse wheel up or down while pressing the **Alt** keyboard key. Pressing **Shift** while wheeling up and down makes the zoom act 10x faster.
Pan	Left-click and drag on the viewport.
Pan all viewports at the same time (Gang Pan)	Left-click and drag on any viewport while holding down the **Alt** keyboard key.
Reset the viewport transformations to its default values	Select the viewport you want to reset by clicking on it and then press **Ctrl+R**
Reset all the viewport's transformations to their default values	Press **Alt+R**
Show the color value of a pixel on screen	Right-Click on the viewport while pressing the **i** keyboard key ("i" is for information).

So to sum it up…

The Viewport Area is where you will see the Viewports displaying all the information contained in the tracks. You can interact with the Viewports by using the mouse and keyboard and display text information on what you are looking at, including DPX metadata when present.

The Menu Bar

At the top of the Main Window is where the Menu Bar lives, and from it you can access some settings that modify the way the program behaves and the viewports are displayed. From there you can also access other windows (including the Load Window which you already know). Most menu items have a keyboard shortcut that is displayed next to it.

Figure 18 The Menu Bar

We will now break the menus down into their individual sub-items

File Menu

- **Load Sequence** (Ctrl+L): Opens the **Load Window.**

- **Save Session** (Ctrl+S): Allows you to save a complete JefeCheck session including loaded tracks, load time parameters and viewport configurations into a single file which can later be recalled. See Section 4: Other Stuff for information about rendering from JefeCheck.

- **Open Session** (Ctrl+O): Opens a complete JefeCheck session. Any sequence settings and loaded tracks will be unloaded and replaced by the information stored in the session file. See Section 4: Other Stuff for information about sessions.

- **Render**: Opens the Render Dialog. See Section 4: Other Stuff for information about rendering from JefeCheck.

- **Preferences** (Ctrl+P): Opens the Preferences Window. See Section 4: Other Stuff for information about JefeCheck's Preferences.

- **Quit** (Ctrl+Q): Quits JefeCheck.

View Menu

- **Reset Current View** (Ctrl+R): Resets the current viewport's transformations.

- **Reset All Views** (Alt+R): Resets all the viewports' transformations.

- **Fullscreen** (Ctrl+F): Toggles full screen mode on or off. If you have two monitors, toggling full screen mode will go from windowed mode, to full screen on one monitor, to full screen across both monitors.

- **Layout**: Allows you to change the layout of the viewports (see Viewport Controls p18).

Color Menu

- **Adjust Screen Gamma** (Ctrl+G): Opens the soon to be gone Adjust Screen Gamma Window. This window allows you to change the gamma of your whole screen. It is not recommended that you use this method and adjust your screen gamma from within your display driver's configuration.

- **BG Color**: Allows you to choose between a Black Background color and an 18% gray Background. This color affects both the Viewports Area background and the Control and Menu bars.

Image Menu

- **Filtering**: When you zoom in and out of a viewport, JefeCheck applies a filter to the displayed image in order to avoid pixilation. You can choose if you want the filter to be a Point Filter (You can see every pixel when you zoom in), or a Bilinear Filter that smoothes out the image. You can control this individually for when you zoom in (Magnification) or for when you zoom out (Minification).

- **FX Controls** (F8): Opens the **FX Control Window**. See Section 2: Processing for more information.

- **FX Manager** (F9): Opens **the FX Manager Window**. See Section 2: Processing for more information.

- **Cube Manager** (F10): Opens the **Cube Manager Window**. See Section 2: Processing for more information.

Remote Session Menu

- **Session Manager** (F6): Opens the **Session Manager Window**. See Section 3: Share for more information.

- **Chat Mode** (Ctrl+Y): Go into Chat Mode. See Section 3: Share for more information.

- **Paste into chat** (Ctrl+V): Pastes text in the clipboard to the remote chat. See Section 3: Share for more information.

- **Save Chat** : Allows you to save the chat log from a Remote Session. See Section 3: Share for more information.

Help Menu

- **Help** (F1): Displays this user manual.

- **About JefeCheck** (F1): Shows the About JefeCheck Window that contains version and contact information and development and testing credits.

PROCESS

2

SECTION 2: PROCESS

In this section you will how to load and use LUTs and JefeCheck FXs.

One of JefeCheck's most powerful features is the ability to process the image sequences you are watching with a variety of Lookup Tables and other effects. JefeCheck uses a simple plug-in system to control how the images are processed.

Real Time Modifications vs Load Time Modifications

As mentioned in Section 1, there are two ways to process an image in JefeCheck, at Load Time and in Real Time. Continuing the film analogy, tracks are like film reels, and viewports are a combination of projectors and projection screens. You can **think of Load Time Processing as making an different print of the film reel**, maybe you crop the image, maybe you print it at a different size, or use different color timings… the point is, no matter how much you adjust the projector lenses or filters, the images on that different print will be changed, and what you see up on the screen won't go back until you reload the original print on the projector again.

In contrast, **Real Time Modifications are equivalent to simply loading the original film reel on the projector, and then putting different filters or lenses on it.** You don't have to make a different print of the reel in order to see a different effect, you simply swap one filter for another, but the reel remains the same.

In JefeCheck, Load Time Modifications are done in the Load Window, where you set how you to load the sequence. Loading a sequence and modifying it at load time is a slower process than simply loading the original sequence, the same way it takes time to make a different print of a film reel.

Real Time modifications on the other hand, are applied to the image on the fly, using the power of your computer's graphics card.

Both Real Time and Load Time modifications can make use of Lookup Tables (LUTs), either 1D (every color channel changes independently and in the same way) or 3D (the way color channels change also depends on the other two).

JefeCheck can load a variety of Lookup Tables formats but the easiest way to get a 3D LUT is to make your own. It is really easy to emulate any color look by using a custom made JefeCheck LUT. If you can render a single frame through your color pipeline, than you can use the color output of that pipeline in JefeCheck! All you do is process the provided calibration image (JefeCheckLUTSourceImage.tga) through your pipeline, the resulting image (should be the same size as the original) is loaded into JefeCheck and converted to a 3D Lookup table that will emulate whatever color alterations your pipeline made to the original image. See Section 4: Other Stuff to find out how to make your own 3D LUT for JefeCheck.

You can also load and use 1D LUTs. The file format for a 1D LUT is a very simple and straight forward text file. To learn how to write your own 1D LUT see Section 4: Other Stuff.

To load a LUT, be it 3D or 1D, you use the LUT Manager.

The LUT Manager Window

From the LUT Manager Window you can load and unload LUTs. To open the LUT Manager Window go to the Menu Bar and from the Image Submenu choose LUT Manager (the notation for that kind of actions will be **Menu>Image>LUT Manager** from now on), or press the **F10 Key**.

The first time you open the LUT manager, it will be empty, so you need to load whatever LUTs you intend to use. To load a LUT you click on the Browse button, this will open a file browser where you can select any of the supported LUT formats that JefeCheck can load (.cub, .cube, .lut, .jlut, .tga, .bmp, .tif). For more information on the formats that JefeCheck supports, see Section 4: Other Stuff. The LUT browser navigates by default to the FX folder in your JefeCheck installation, and that is where your LUTs should be stored for easy access.

Once you have selected a LUT file, click OK in the browser and JefeCheck will start loading it. If the LUT was correctly loaded, the progress bar under the Browse button will turn green and notify you of the success.

The LUT will appear in the Loaded LUTs section of the LUT Manager Window. You don't have to load a specific LUT every time you need it, you can tell JefeCheck to auto-load it for you whenever you open up the program. Simply check the Auto-Load checkbox next to the loaded LUT, and JefeCheck will automatically load it next tieme.

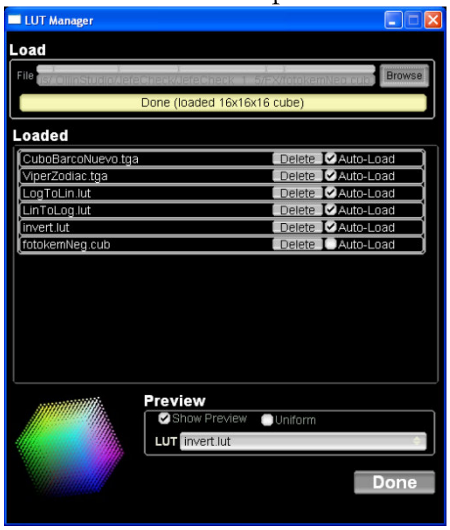


Figure 19 The LUT Manager

You can load as many LUTs in different formats as you want. You unload them by pressing the Delete button next to each one. This only unloads them and does not delete them from disk.

Visualizing LUTs

JefeCheck can show you a visual representation of the LUT you have loaded, be them 1D or 3D. To visualize a LUT you click the Show **Preview Checkbox** un the Preview Section of the LUT Manager Window and select the LUT you want to visualize from the **LUT option box**. You will see the LUT drawn on a Viewport in the Main Window. If you have more than one viewport in the layout, you will see the LUT on each one, and you can handle each instance independently. You use the same controls you normally use to pan and zoom a viewport to handle the LUT visualization (see Viewport Interaction p25).

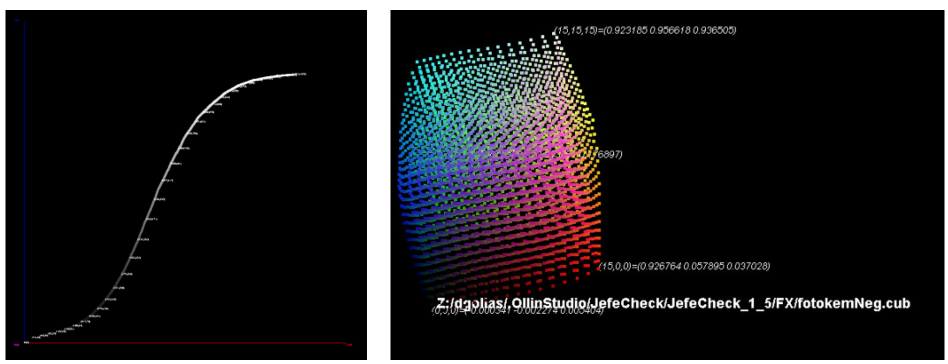


Figure 20 Examples of 1D and 3D LUT visualization

1D LUTs are plotted on an *x,y* plane, with the input value on the X axis, and the output value on the Y axis. A grayscale value of the output is used to color each point in the plot. A text label of a sub sample of the values is also drawn showing the input and output values.

3D LUTs are displayed as color cubes, placing a colored point at each sample in 3D space. A canonic 3D LUT would generate a perfect color cube, while a modified cube would display other characteristics. Each point in the LUT is positioned in a point in space corresponding to its output value in RGB, and is colored with that same information. This will deform the cube showing distinctive shapes for each LUT in a way that an experienced colorist will be able to predict the kind of color transformation that each cube will yield. If you click the Uniform checkbox, the samples in the cube will be spaced uniformly and form a perfect cube, but the points will still be colored with their output values. This way you can compare two cubes without being distracted by the geometric deformations and only focus on the color changes.

Using LUTs

Of course, the reason you load a LUT is not only to visualize it, but to actually use it to transform the colors in an image. As you may remember from Section 1, you can use one or more LUTs (or none) at load time, stacking one on top of the other. The LUTs you have loaded will be available from the LUTs option box in the Load Window.

Although Appling LUTs at load time is OK for lower end computers, is your system has a decent video card, the best option is to use Real Time processing to apply LUTs and other effects to the image.

Real Time modifications are called FXs within JefeCheck. Each FX comes in the form of a plug-in that must be loaded in order to be used. JefeCheck comes with two FXs for LUT application, one for 1D LUTs and one for 3D LUTs. See appendix A for an explanation of the included FXs.

To load an FX you use the FX Manager in a similar way to how you use the LUT Manager to load LUTs.

The FX Manager Window

From the FX Manager Window you can load and unload FXs. To open the FX Manager Window go to the Menu Bar and from the Image Submenu choose FX Manager (the notation for that kind of actions will be **Menu>Image>FX Manager** from now on), or press the **F9 Key**.

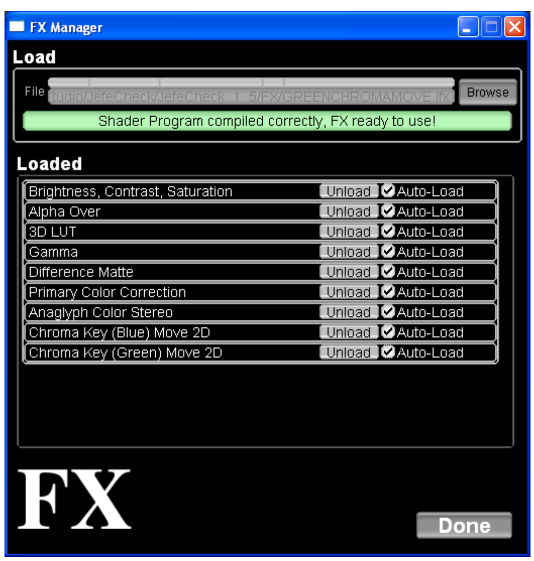


Figure 21 The FX Manager Window

When you first open the FX Manager Window, it will be empty. To load an FX, click the Browse Button. This will open a file browser window in the default FX folder within your JefeCheck installation. You can navigate to other folders if your FXs are stored somewhere else. Select one of the .jfx files and click ok to start loading the FX. If everything goes OK you will see a confirmation message in the progress bar under the Browse Button, the FX will now be loaded and displayed in the Loaded section of the FX Manager. Load as many FXs as you want. **You don't have to load the FXs by hand each time you open JefeCheck, to have the FXs load automatically check the**

Auto-Load Checkbox next to each one to have them load next time you use JefeCheck, as shown in Figure 21.**Error! Reference source not found.**. To unload n FX click on it's unload button. Once an FX is loaded it can be used on a Viewport to modify the image.

The FX Control Window

You use FXs by applying them to viewports, a viewport can have no FX applied or as many as you graphics card performance will allow you while maintaining a desired playback frame rate. **To apply an FX you use the FX Control Window**. To bring it up, click on **Menu>Image>FX Control** or press **F8**. You can also click on the FX Button on the Main Window's Control Bar.

Remember that FXs are applied to each viewport independently, this way you can have one viewport playing back the unprocessed sequence, while on the viewport next to it, you can have the sequence playing back with a LUT or Primary Color Correction applied.

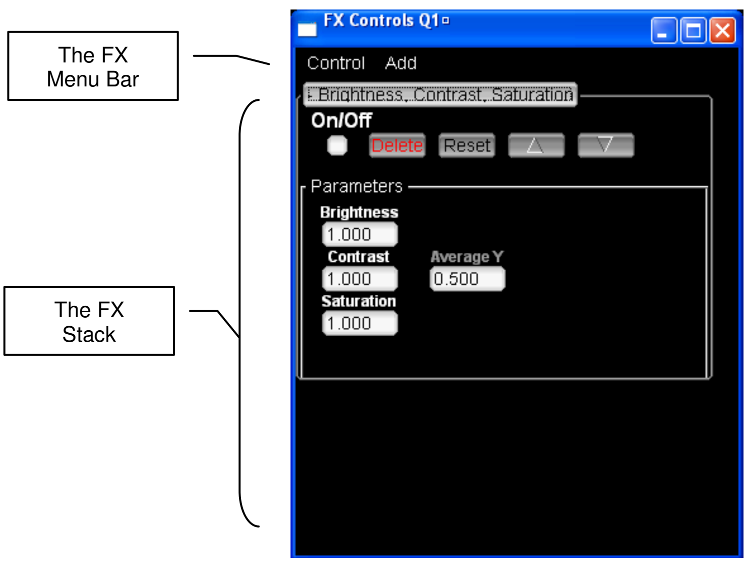


Figure 22 FX Control Windows with a Brightness, Contrast, Saturation FX applied

The FX Control window has its own menu bar from which you can access the FX Manager and apply FXs among other things (see FX Control Window Menu p34).

To apply an FX to a Viewport, open up the FX Control Bar (F8) and click on the Viewport you wish to use, the FX Control Window title will change to show what viewport is selected (e.g. *FX Controls V1* means you are working on Viewport 1), if not already there, an **Add** submenu will appear on the window's menu bar from which you can select what FX you wish to apply. To switch viewports in order to add, remove or modify FXs on them, simply click on the viewport which you wish to become active. The FX Control Window will change the title to show what Viewport you are working on.

Once you click on an FX from the Add menu, it will appear on the FX Stack as shown in Figure 22. The FXs are applied to the image in the order in which they were added to the stack, so the outcome of an FX will be passed to the one under it.

Common FX Controls

All FXs have a set of common controls:

- **Name Bar** 

: Shows the name of the FX and allows you to collapse all the controls to save space in the FX Control Window.

- **On/Off**: Turns the FX on or off. Once this checkbox is enabled, you will see the effect of the FX on the viewport. This way you can see the effect of an FX without having to delete it and re-apply it.

- **Reset**: Returns all the FXs parameters to their default value.

- **Move Up**: Moves the FXs up the stack, meaning it will be applied before the one under it.

- **Move Down**: Moves the FXs up the stack, meaning it will be applied before the one under it.

Most FXs will have parameters that you can set to customize their behavior. For example, the **Brightness, Contrast, Saturation** FX has controls for brightness, contrast, saturation and average luminance. Every FX's controls can be different, but they share some common characteristics. As with almost all other numeric input fields in JefeCheck, an FX's numeric parameters can be modified in two ways: by typing in a number, or by clicking and dragging with a mouse button on the field. The value will change according to the mouse movement at a rate dependant on what mouse button you click. If you click and drag with the left mouse button, the value will vary at the smallest decimal precision allowed by the field (if the field is something like Brightness then your mouse movements will increment or decrement the value by 0.001), if you middle click, the value will move at a 10x rate, and if you right click, it will move at a 100x rate.

Performance

A Viewport can have many FXs stacked one on top of the other, (although not all need to be on at the same time). You can apply an arbitrary number of FXs to a stack, but the performance and frame rate of the real time playback with many FXs applied will vary depending on your graphics hardware. When shopping for graphics cards, the main feature you should keep an eye for is the amount of Pixel Shaders or Pixel units (called Stream Processors in the newest graphics cards), the more of these that a graphics card has, the better the performance of an FX stack will be. At the time of writing, video cards that comply with the minimum hardware requirements have anywhere between 12 and 128 pixel shaders. The FX performance difference from these cards is not noticeable when a single simple FX is applied, but as the number of FXs grows, the performance at the lower end will suffer.

The amount of work that an FX can demand from a graphics card varies depending on the complexity of the FX, but tests have shown very good performance on mid range gaming level video cards with an excess of 5 FXs applied on a 2K resolution sequence.

FX Control Window Menu

The FX Control Window has its own menu bar which consists of two menus, the add menu used to add FXs to a stack (when FXs are available), and a Control Menu with the following items:

- **FX Manager**: Opens up the FX Manager Window if you need to load an FX.

- **Clear All**: Removes all FXs from the stack (equivalent to pressing the delete button on each FX).

- **Save Stack**: You can save a stack with all the FXs and each one with their own particular settings into a file. You can later apply this same stack to another viewport or a different sequence. When you click on this, a browser window will appear, allowing you to choose a location and type in a name for the stack. JefeCheck FX stacks have a .fxs extension.

- **Load Stack**: If you have previously loaded an FX Stack, you can load it by selecting this item. Select the .fxs file from wherever it was stored and JefeCheck will place the appropriate FXs with the saved parameters on the stack. The FXs need to be loaded for the Load Stack function to work; otherwise you will get a message stating what FX is missing. Also note that when loading a Stack, the FXs contained there will be added to the current stack. If you want to replace the current stack, just clear all the applied FXs prior to loading.

SHARE

3

SECTION 3: SHARE

In this section you will learn how to start and participate in a JefeCheck Remote Session.

In the previous sections you learned how to use JefeCheck to play sequences and process them using FXs, but the last big feature of JefeCheck is the ability to do all that in collaboration with people that might be miles away.

Collaborating with other people over a network connection or the Internet is called a Remote Session in JefeCheck.

Remote Sessions

When two or more people have a JefeCheck Remote Session, each participant has a copy of JefeCheck running on their computer, along with a local copy of the sequences that will be loaded for playback. JefeCheck does not send any image information over the network, this is for two reasons:

1. Sending image information requires great amounts of bandwidth that might not be available where a Remote Session participant might be.

2. Even if everybody had the bandwidth to send image data in real time, sensitive content such as frames from an unreleased movie are usually copied and sent

on a need to have basis through specialized, controlled and secured channels, managed by a data wrangler.

A JefeCheck Session requires at least two participants, one of who will need to act as a Session Server. If you have ever played a networked video game then you already know what this means, but if you haven't, the explanation is very simple and can be summed in three steps:

1. **Start a Server**

   One participant, usually the one with the fastest connection and best computer starts a server from JefeCheck. This participant will need to have an IP Address that other players can access from their network, normally called a Public Address if the session is being held over the Internet.

2. **Other Participants Connect to the Server**

   Once the server is started, the server communicates to all other participants that the server is up and running, along with the IP Address that they must use to connect to it.

3. **Play!**

   When everybody has connected to the server, they will all be in the same session, meaning that anything someone does on their copy JefeCheck will be mirrored on everybody else's copy, including the server.

Everything involved in setting up or joining a Remote Session is done from the Remote Session Manager Window.

The Remote Session Manager

To start a remote session or to join one, you use the Remote Session Manager Window. To bring it up you go to **Menu>Remote Session>Session Manager** or press F6 as seen in Figure 23.

Connecting to a Server

The first section in the Session Manager Window is Connect to Server. In order to connect to a JefeCheck remote server that's already online you do the following:

1. **Type the IP address in the IP text box** (or select a recent one from the drop down menu next to it).

2. **Type in the port in the Port text box** for the server we are connecting to (ports 32000 and above recommended).

3. **Type in the password in the Password box** (if the server requires one).

4. **Type in a Nickname in the Nickname box**, other participants in the session will identify your actions with this nickname.

5. **Click the Connect Button**.

The status box will indicate if the connection to the server was successful. If you are online with the server, the status box will turn green and say **Connected to server**.

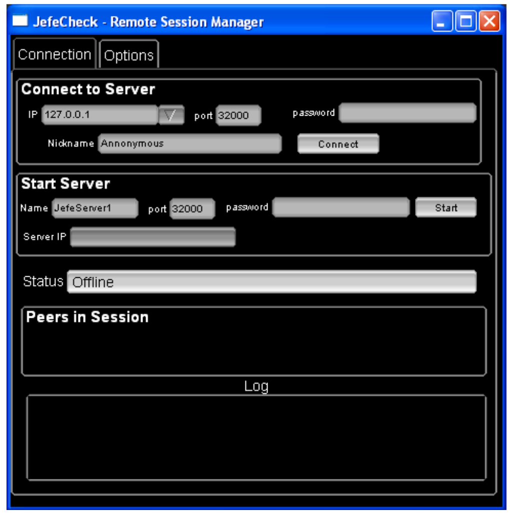


Figure 23 The Remote Session Manager Window

Starting a Server

The second Section in the Remote session manager is the **Start Server** section. You use this to start a server in order to host a remote session that other users can log into. When you are a server you are also a participant in the session as if you had connected to a server yourself. To start a server you do the following:

1. **Type the Name of the server in the Name box**, the server's name will be your nickname.

2. **Type in the port in the Port text box**, other users will use this port to connect to you (ports 32000 and above recommended).

3. **Type in the password in the Password box** (if you wish to require a password to connect to your server, recommended).

4. **Click the Start Button**.

If starting the server is successful, the status bar will turn green and say **Server Online**. Also, the Server IP text box will show the IP address that other participants should use when connecting to you, along with the port that you are using (e.g. 172.16.28.105:32000, IP address 172.16.28.105 on port 32000), you should communicate this information to others attempting to connect to you.

Participating in a Remote Session

Once a server is set up and at least one more user is connected to the server, you can start using JefeCheck the way you normally would.

Loading Sequences in a Remote Session

The only thing not handled automatically for you when you have a remote session is loading. Users should communicate to each other what image sequence (or sequences) will be loaded for review, and in what track.

This is because users may have different file system organizations; maybe one participant is loading the images from a folder on a CD, another one from a shared folder on a network, and a third one is loading from a local folder on a hard drive. Each user is responsible for loading the correct sequences into the correct tracks.

One important thing to note is that although all users MUST have the same image sequences loaded, they don't have to load them at the same scale. That is, a user with a laptop can load the sequence at 50% or 25%, and a user on a bigger workstation can load the sequence at full resolution. JefeCheck will be aware of this difference in size as long load the same files and do the scaling from the Load Window (see **Load Time Modification Controls**:Scale p8).

If you know how to use JefeCheck then you know how to participate in a Remote Session, you can do everything you do in JefeCheck normally and all other participants will see the changes reflected on their own JefeCheck. So you can pan, zoom, change layouts, play, scrub the timeline, advance or rewind frames, change In and Out points of the timeline, change the frame rate, change the playback mode, add and modify FXs (apply LUTs, Primary Color correction, mix two sequences etc) flip, flop… you get the idea.

There are only a few new commands involved in using JefeCheck in a Remote Session: Chat and Remote Pointers.

Chat

In JefeCheck you can text chat with all other participants, meaning you can send text messages that all participants will see on their screen while you interact with the rest of JefeCheck.

Figure 24 Chat Interface

When you are connected to a JefeCheck Remote Session press **ctrl+y** to bring up the Chat Interface. Press **Escape** to exit the chat interface. The Chat interface is overlaid at the bottom of the Viewport Area of the Main Window as seen in Figure 24. The Chat interface is very simply and you will understand it instantly if you have ever used an instant messenger program. There are two parts, divided by a horizontal line, when you have are in chat mode, you type messages to the lower part simply by typing it on your keyboard and then press enter to send.

Whenever you receive a message, you will see it above the line that divides the received messages and the area where you type your messages.

You will only see the last 5 messages received, if you want to go back to see older messages, make sure you are in chat mode and press the **Up** or **Down** cursor keys on your keyboard to navigate through messages.

Remember you exit the Chat mode by pressing the **Escape key**.

When you are not in chat mode, you will still be able to see the messages sent to you, to hide them press the **c key** on your keyboard. Otherwise, unless you are in chat mode, sent messages will fade away after a few seconds.

You can also save the whole Chat in a session by going to **Menu>Remote Session>Save Chat**, you will see a file chooser dialog, simply navigate to the path where you want to save and type in a name for the file. This is saved as a plain text file.

Remote Pointers

Asides from text, you can also make use of JefeCheck's remote pointers. This is a way to make everybody in the session see exactly where you are pointing at on the screen with your mouse.

To use a remote pointer, you simple click with your mouse on the Viewport Area (with any mouse button). Everybody else in the session will see a white point with your nickname next to it. As long as you keep any button pressed on your mouse, the others will see your remote pointer. You can move your mouse while you keep the mouse button pressed and the others will see your remote cursor moving. When you left-click and drag to pan the image, the others will also see your remote cursor moving the image.

Note that if both users loaded the same sequence at different scales using the Scale parameter in the Load Window, JefeCheck's remote pointers will make up for the difference in scales. So you know that when you are pointing at a specific point in your viewport, the others are seeing your remote pointer at that same place in the image. This is also true for different monitor resolutions. Remote pointers work by showing where you are pointing at on the image on screen, not at where you are pointing at on your monitor.

Using both the chat and the Remote Pointers you can get some very useful interaction between all the participants in the session. Combining that with FXs to change colors or bring up difference mattes or anything else you can think of makes JefeCheck a very powerful remote review tool.

OTHER
STUFF
4

SECTION 4: OTHER STUFF

This section is not yet ready for the Beta Demo Version.

Work in progress. Look back in this section in the future.

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